

MPC-MAP Assignment No. 4 - Report

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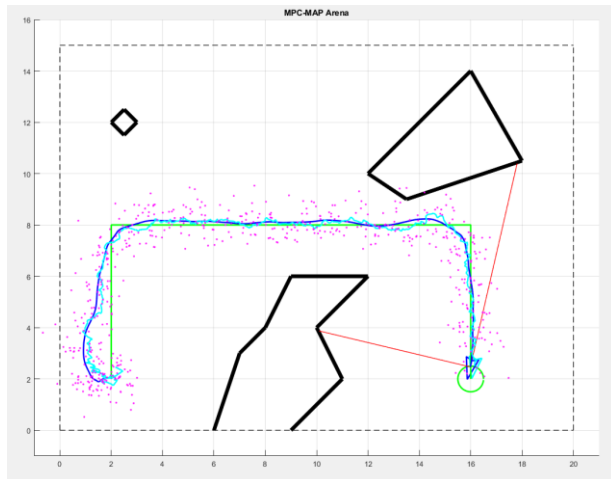
Task 1

I loaded the outdoor_1 map and manually routed a path via $y = 8$ to avoid obstacles. The robot stands still for the first 50 iterations to collect GNSS data, calculating its initial position and the measurement covariance matrix $Q = 0.28$.

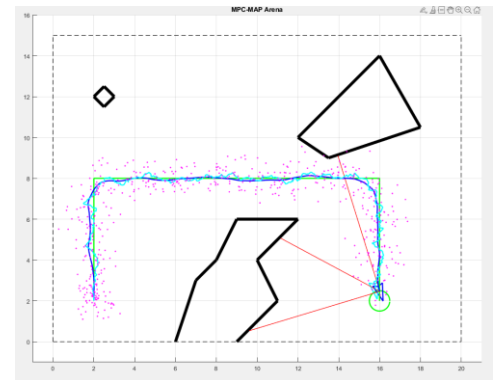
Task 2

For prediction (ekf_predict), I calculated the Jacobian matrix G to linearize the non-linear motion model. For correction (kf_measure), I used a standard linear Kalman Filter (matrix H) since GNSS provides direct X and Y measurements.

Task 3



Starting with a known pose and zero covariance, I used only the EKF estimate for control. I tuned the process noise R to $\text{diag}([0.001, 0.001, 0.001])$, trusting the smooth kinematics over noisy GNSS. I also added a spin-in-place logic to prevent wall collisions.



Task 4

To simulate a blind start, I initialized the heading belief to 0 with a high variance (100). The robot initially made a 0.75-unit arc due to the wrong heading, but quickly converged as the EKF processed GNSS data. To eliminate oscillations after convergence, I further lowered R to $\text{diag}([0.00055, 0.00055, 0.00055])$ and the controller gain to $K_p = 1.2$, ensuring a perfectly smooth ride.

